

Estimation of Required Horizontal FoV for Ideal HMD Utilizing Vignetting under Practical Range of Eye Displacement

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ABSTRACT

This study presents the required horizontal FoV for ideal HMDs that bring an indistinguishable experience from the state of not wearing them. We investigated the threshold of the horizontally visible range relative to the head, when the size of the eye displacement is equivalent to that in real life. We confirmed that the required FoV was smaller than that in a previous study, in which the participants' eye displacement was at its maximum. We found that the required FoV was approximately 229° when the peripheral luminance gradually decreased, as in vignetting.

Keywords: Field of view, HMD, eye-head coordination.

Index Terms: Human-centered computing—Visualization—Empirical studies in visualization

1 INTRODUCTION

The goal of head-mounted displays (HMDs) is to provide an experience in which users cannot distinguish between the virtual and real world. Such HMDs, referred to as ideal HMDs in this study, must offer sufficient capabilities in various features such as field of view (FoV), spatial resolution, or temporal resolution, which are indistinguishable from observing the world without HMDs. Revealing the thresholds of these features helps in designing a future virtual reality system. As a first step, we investigated the horizontal FoV required for ideal HMDs.

Because an HMD is fixed to a user's head, the FoV required for ideal HMDs is the entire visible range based on the head while allowing eye movements. Typical studies of the FoV have focused on the visible range with steady fixation of gaze toward the front target, as shown in Fig. 1(a), and have reported that the limitation of the FoV on the temporal side was up to approximately 110° for monocular viewing [1], [2]. However, the eye-in-head displacements expand the visible range, as shown in Fig. 1(b). Consequently, ideal HMDs should cover the horizontally visible range, originating at the head instead of the eyes (Fig. 1(c)).

A previous study investigated the required FoV by a target detection task under conservative conditions [3]. Participants maximally moved their eyes toward the direction of a target stimulus, an LED point light, while their head was fixed at a chin rest. It revealed that the required horizontal FoV was approximately 253° for binocular viewing, suggesting that the combination of the FoV and eye displacement, shown in Fig. 1(b), overestimates the required FoV. For example, if the eye displacement is 35° , the estimated FoV should be 290° , which is significantly larger than the FoV of 253° .

Here, we assumed more practical conditions that

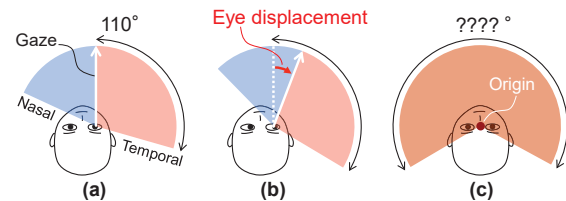


Figure 1: Horizontally visible range when gazing (a) frontward and (b) rightward for monocular viewing, and (c) both leftward and rightward for binocular viewing.

correspond to actual usage, where the eyes are expected to rarely move to their maximum while the head moves freely. When gazing at an object distant from the frontal direction, the head and eyes cooperatively move toward the object, known as *eye-head coordination*, which reduces eye displacement [4]. We expect that this reduction in eye displacement alleviates the required FoV in actual usage.

Additionally, the point light stimulus in the previous study may have been excessively conspicuous. The human visual system is more sensitive to steep variations in luminance in space than gradual variations [5]. Notably, actual HMDs can gradually reduce the luminance of the peripheral region as *vignetting*, which probably makes the FoV limitation less noticeable.

We investigated the required horizontal FoV while assuming actual usage. In the experiment, participants moved both their eyes and heads naturally orienting toward a target. We used natural scenes instead of the point light and decreased the peripheral luminance *steeply or gradually*.

2 METHODS

We recruited 16 participants (8 male, 8 female) with a visual acuity of at least 20/23 without glasses. They provided written informed consent, and the study was approved by the Ethics Committee of our laboratories.

We used a cylindrical display that simulated the horizontal FoV of HMDs (Fig. 2(a)), consisting of 20 4 K panels (LG, 65EV5E) curved along with the cylindrical form. We obtained the position and posture of the head using nine motion trackers (OptiTrack, Prime X13) that tracked the retroreflective markers attached to a helmet mounted on the head. To simulate the FoV of actual HMDs, we blacked out the *limited region*, the area more peripheral than the border, and displayed the image of the scenes in the *displayed region*, the area more central than the border. The displayed region moved according to the direction of the head (Fig. 2(b)). Although this apparatus did not fully simulate actual HMDs, e.g., lack of wide vertical FoV, binocular disparities, and 6 degrees of freedom, it simulated the horizontally limited FoV of actual HMDs based on head orientation, which seemed to function successfully as a substitute for HMDs.

We prepared three virtually natural scenes that had

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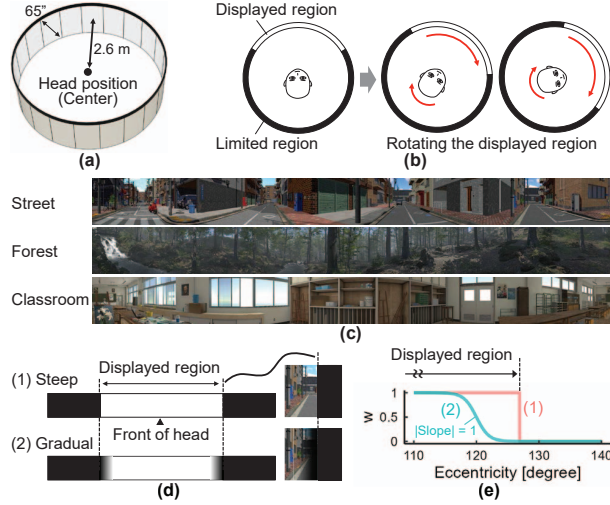


Figure 2: The experimental setup. We used (a) the cylindrical display, (b) which simulated the limited FoV of HMDs. We showed (c) three virtual scenes that peripheral luminance decreased (d) steeply or gradually (e) based on the step or sigmoid functions.

360° horizontal FoV (Fig. 2(c)). We rendered images of these scenes from 3-D models as projections onto the cylindrical display.

We investigated two conditions: steep and gradual variation in luminance over the borders between the displayed and limited regions (Fig. 2(d)). We blended pixel values as $p = w p_{org} + (1 - w) p_{black}$, where p_{org} and p_{black} represent the pixel values of the original and black images, respectively, and w is the weight ranging from 0 to 1. As shown in Fig. 2(e), w changed against eccentricity, angular distance from the central point, by the step and sigmoid functions for the steep and gradual conditions, respectively. The same number of participants, eight, were assigned to each condition.

We manipulated the size of the displayed region in 11 levels with an interval of 5°: 225–275° for the steep condition and 238.8–288.8°, for the gradual condition. Each level of the stimulus was presented 10 times.

The participants sequentially observed two images of the same scene; one had the displayed and limited regions, and the other had only the displayed region of 360°. At every start of each interval, the head positions and postures were adjusted. Next, either image was shown with a fixation circle that indicated the initial frontal direction. After 0.5 sec, a 1° bullseye target appeared at 60° rightward from the fixation circle, which disappeared when the target appeared. The participants directed their eyes and heads toward the target naturally, as in real life. The scene image turned black 1.5 sec after the appearance of the target. Similarly, a second interval was performed. After the two intervals, the participants were asked to report which interval contained a limited region.

In the analysis, we estimated the threshold FoV with which the presence or absence of the limited region was just indistinguishable. We assumed the following model consisting of a cumulative distribution function (CDF) for normal distributions: $p_i(\text{Correct}|x_i, \mu, \sigma) = 1 - \text{NormalCDF}(x_i, \mu, \sigma)/2$, where the correct responses were obtained with probabilities p_i when given the displayed region x_i . We estimated the mean,

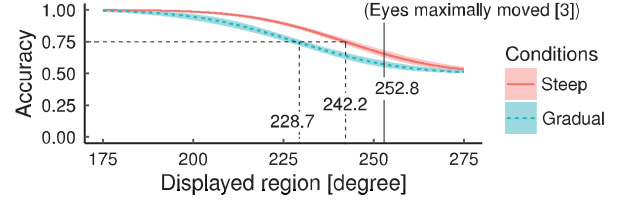


Figure 3: Predicted accuracies for the displayed region. The accuracy of 0.75 indicates the threshold for just not noticing the limited FoV. Each error band indicates the 95% prediction interval.

μ , and standard deviation (SD), σ , of the normal distribution by Bayesian estimation, using Markov chain Monte Carlo method. We removed data from an individual in the gradual condition because of low accuracy, which is almost equivalent to the chance level of 0.5.

Before the estimation, we performed post-hoc calibration for motion-to-photon latency, from head movement to drawing, because the latency narrows the substantial displayed region. The mean latency was approximately 136 ms (SD, 6 ms). With the latency and recorded head motions, we obtained the maximum difference in angles d_{max} between the head orientation and center of the displayed region for each trial. Then, the size of the displayed region x_i was corrected to $x'_i = x_i - 2d_{max}$.

3 RESULTS

With the practical range of eye displacements, the required FoV was smaller than that with the maximum eye displacements in the previous study [3] (see Fig. 3). In detail, the threshold for the steep condition was 242.2° with a 95% confidence interval (CI) of 239.9–244.8° while the required FoV in the previous study was about 252.8°.

The required FoV in the gradual condition was alleviated compared to that in the steep condition. Specifically, the threshold for the gradual condition was approximately 228.7° with a 95% CI of 226.0–231.2°, not overlapping the 95% CI of the steep one.

4 CONCLUSION

We investigated the required horizontal FoV for ideal HMDs assuming actual usage. We found that a horizontal FoV of approximately 242° was sufficient when the eyes and head moved naturally; the FoV was smaller than that with the maximum eye displacements [3]. Additionally, the required horizontal FoV was further reduced to 229° by daring to degrade the display quality with vignetting-like luminance variation. We believe that our findings will be helpful in establishing a design for ideal HMDs.

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